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EVEN END SEMESTER EXAMINATION

2025

Name of the Course: B.Tech
Civil Engineering

Semester: VI Paper Code: TCE 604

Name of the Paper : Quantity
Estimation and Costing

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1

(20marks) CO1

- (a) Calculate the cost by using a preliminary estimate of a polytechnic building for 800 students in order to assess the amount of fund on the following particulars:
 - 1) Carpet area required per student = 1.20 Sq.m
 - 2) Area of corridor, verandah and lavatories etc = 25 % of the plinth area.
 - 3) Area of walls = 15% of the plinth area
 - 4) Considered plinth area rate = Rs. 1545.00 per sq.m
 - 5) Cost of water supply = 6% of building cost.
 - 6) Cost of sanitation = 5% of building cost.
 - 7) Cost of electrification = 12% of the building cost.
 - 8) Cost of approach road and boundary wall = 2.5% of the building cost.
 - 9) Contingency and work charged establishment shall be 3% and 2.5% of the total cost
- (b) Calculate the rough estimate for a proposed commercial complex for a municipal corporation for the following data.

Plinth Area = 550m²/ floor
Height of each storey 3.5m
No. of storey = G+7
Cubical content rate = Rs. 1550/m³
Assume all suitable charges ?
- (c) Explain in detail measurement sheet and abstract sheet?

Q2

(20 marks) CO2

- (a) Calculate the quantity of material and labour for RCC (1:1.5:3) in beam including centering and shuttering? Assume all suitable data take appropriate units ?
- (b) Calculate the quantity of material and labour for traditional brick of (25.4cm X 12.7 cm X 7.6 cm) in superstructure? Assume suitable data
- (c) Calculate the quantity of material and labour for 6mm thick mosaic flooring (1:1.5) over cement concrete (1:2:4) for 100 square meter. Assume suitable data?

Q3

(20 marks) CO3

- Explain the Long Wall Short Wall and Center Line method with their priorities?
- (a)
 - (b) Explain mid-sectional area method and mean sectional area method for earthwork in roads?
 - (c) Calculate the cost of earth work for a portion of road for 400 m length from the following data: -

Formation width of road is 10m. Side Slopes are 2:1 in banking and 1½: 1 in cutting. Adopt suitable rates

Station	Distance in metre	R.L. of Ground	R.L. of formation
25	1000	51.00	52.00
26	1040	50.90	
27	1080	50.50	
28	1120	50.80	
29	1160	50.60	
30	1200	50.70	
31	1240	51.20	
32	1280	51.40	
33	1320	51.30	
34	1360	51.00	
35	1400	50.60	

Downward gradient
of 1 in 200

Q4

(20 marks)

- Explain overhead charges and work charge establishment and their importance in civil construction works? CO4
- (a) Explain various percentages for different services in building?
- (c) Explain Cost of Works and describe factors affecting cost of works ?

Q5

(20 marks)

- Explain methods of fixation and calculation of rent of government building in detail ? CO5
- (a) A coloniser intends to purchase a land of 100,0000 sqm area located in the suburb of a big city to develop it into plots of 900sqm each after providing necessary roads and parks and other amenties. The current sale price of small plots in the nieghbourhood is 1550 per sqm. The coloniser wants a net profit of 20 %. Work out the maximum price of land at which the coloniser may purchase the land ?
- (c) Explain depreciation, sinking fund, and scrap value?